# 6-Month Full-Stack Web Development Course

JavaScript • CSS • Core PHP • MySQL • Laravel • Vue.js

**Course Duration:** 6 Months | **Level:** Beginner to Job-Ready

## Course Overview

This 6-month practical course is designed to take students from web-development fundamentals to building complete full-stack applications. Students will learn frontend fundamentals, JavaScript, Core PHP, MySQL, Laravel, and Vue.js, with hands-on projects and a final capstone application.

## Learning Outcomes

- Build responsive web pages using HTML, CSS and modern layout techniques.
- Write JavaScript for dynamic interfaces, validation and API communication.
- Develop server-side applications using Core PHP and MySQL.
- Understand MVC architecture and build REST APIs with Laravel.
- Create modern reactive frontends using Vue.js.
- Connect Vue.js frontends with Laravel APIs.
- Use authentication, validation, CRUD operations and database relationships.
- Build and present a complete full-stack capstone project.

## Month 1 — Web & CSS Fundamentals

- HTML5 essentials: document structure, semantic tags, forms, tables and media.
- CSS fundamentals: selectors, cascade, specificity, box model and positioning.
- Responsive design: Flexbox, Grid, media queries and mobile-first layouts.
- Typography, colors, spacing, buttons, cards, navigation bars and reusable UI patterns.
- Introduction to browser developer tools and debugging.
- Mini Project: Responsive Personal Portfolio / Business Landing Page.

## Month 2 — JavaScript Fundamentals

- JavaScript syntax, variables, data types, operators and control flow.
- Functions, arrays, objects, loops and common array methods.
- DOM manipulation and event handling.
- Form handling and client-side validation.
- ES6+: let/const, arrow functions, destructuring, spread/rest and modules.
- JSON, localStorage and basic asynchronous programming.
- Fetch API and introduction to REST APIs.
- Mini Project: Interactive Task Manager / Student Management UI.

## Month 3 — Core PHP & MySQL

- PHP syntax, variables, conditions, loops, functions and arrays.
- Forms, GET/POST, sessions, cookies and file handling.
- Reusable PHP components and basic application structure.
- MySQL fundamentals: databases, tables, keys and relationships.
- SQL: SELECT, INSERT, UPDATE, DELETE, WHERE, JOIN, GROUP BY and ORDER BY.
- PHP–MySQL integration using PDO and prepared statements.
- CRUD application development and basic security practices.
- Project: PHP + MySQL Employee / Inventory Management System.

## Month 4 — Laravel Framework

- Laravel installation, project structure, configuration and environment setup.
- Routing, controllers, middleware and request handling.
- Blade templates, layouts, components and forms.
- Database migrations, seeders, factories and Eloquent ORM.
- CRUD operations with Eloquent.
- Relationships: one-to-one, one-to-many and many-to-many.
- Validation, authentication, authorization and error handling.
- File uploads, pagination and search/filter functionality.
- REST API development with Laravel.
- Project: Laravel Admin Panel + REST API.

## Month 5 — Vue.js

- Vue.js fundamentals and project setup.
- Components, props, emits and component communication.
- Reactive data, computed properties and watchers.
- Forms, validation, conditional rendering and lists.
- Vue Router for multi-page SPA navigation.
- State management concepts and reusable components.
- Calling REST APIs with Axios / Fetch.
- Authentication flow and protected frontend routes.
- Project: Vue.js Dashboard consuming an API.

## Month 6 — Full-Stack Integration & Capstone

- Laravel API + Vue.js architecture and integration.
- Authentication using API tokens / Sanctum concepts.
- CRUD workflows from Vue frontend to Laravel backend.
- Database relationships and complex API responses.
- Search, filtering, sorting, pagination and file uploads.
- Error handling, validation and secure API practices.
- Git/GitHub workflow, environment configuration and deployment basics.
- Debugging, testing basics and application optimization.
- Final Capstone: Complete Full-Stack Business / Management Application.
- Final presentation, code review and interview preparation.

## Recommended Weekly Structure

| Activity | Recommended Time |
|---|---|
| Theory & concepts | 2 hours/week |
| Live coding / demonstrations | 2 hours/week |
| Hands-on practice | 3 hours/week |
| Assignment / mini project | 2 hours/week |
| Review & assessment | 1 hour/week |

## Assessment & Certification

- Monthly assignments and practical assessments.
- Mini projects after major modules.
- Code quality, database design and API integration reviews.
- Final full-stack capstone project and presentation.
- Course completion certificate based on attendance, assignments and final project.

## Technology Stack

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|---|---|
| **Frontend** | HTML5, CSS3, JavaScript, Vue.js |
| **Backend** | Core PHP, Laravel |
| **Database** | MySQL |
| **API** | REST API, JSON, Axios / Fetch |
| **Tools** | VS Code, Git, GitHub, Browser DevTools |

**Final Goal:** By the end of the course, students should be able to design, develop, connect, test and present a production-style web application using Laravel, MySQL and Vue.js.